

Kinematics of Simple Harmonic Motion

Defining SHM:

- Restoring force $F = -kx$ (where k is spring constant)
- $x = A \sin(\omega t + \phi)$
 - x = displacement, A = amplitude,
 - ω = angular frequency, ϕ = phase constant

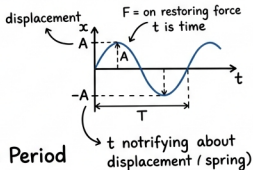


Time Period:

- Angular frequency $\omega = \frac{2\pi}{T}$

$$\Rightarrow \omega = \frac{2\pi}{T} \Rightarrow 2\pi \sqrt{\frac{m}{k}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{k}} \quad \begin{array}{l} \rightarrow T \text{ is Time Period} \\ m \text{ is mass} \end{array}$$



$$\omega t = \frac{2\pi - A \sin(\omega t + \phi)}{T}$$

Velocity & Acceleration:

- $v = \omega \sqrt{A^2 - x^2} \rightarrow$ Maximum velocity $v_{\max} = \omega A$ at $x=0$
 - $\hookrightarrow$ Maximum angular frequency in displacement
- $a = -\omega^2 x \rightarrow$ Maximum acceleration $a_{\max} = \omega^2 A$ at $x=A$
 - $\downarrow$
spring velocity

Energy in Simple Harmonic Motion & Graphical Analysis

Energy in Simple Harmonic Motion (SHM)

Kinetic energy is a spring constant, to stround amplituy and potential energy of the harmonic motion.

Kinetic Energy: $KE = \frac{1}{2} k (A^2 - x^2) \Rightarrow \frac{1}{2} k (A^2 - x^2)$
 $= \frac{1}{2} k (A^2 - x^2) = \frac{1}{2} (A^2 - x^2) = \frac{1}{2} k A^2$

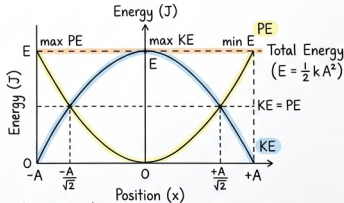
Potential Energy: $PE = \frac{1}{2} k x^2$

k = spring constant

A = amplitude

x = displacement

Total Energy: $E = \frac{1}{2} k A^2$



At $x = 0$ (equilibrium),
 $PE = 0$, KE is max (E)

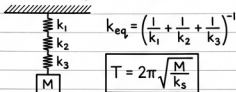
At $x = \pm A$ (extremities),
 $KE = 0$, PE is max (E)

The low energy formula's the m-KE or tind +k E is max (E).
 Total Energy contretions in amplitude, its given at the
 min = KE enenty $\pm E$ is +A.

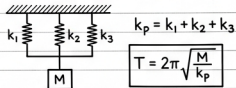
Pendulums and Spring Systems

Spring Systems

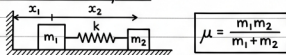
Series Spring-Mass System



Parallel Spring-Mass System

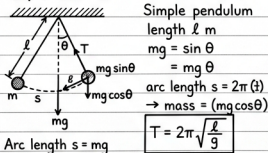


Reduced Mass System

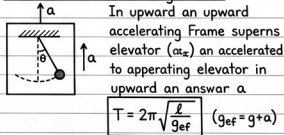


Pendulum Systems

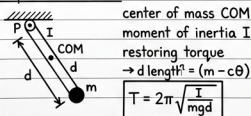
Simple Pendulum



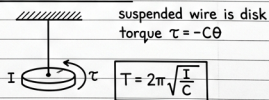
Pendulum in Accelerating Frame



Compound Pendulum



Torsional Pendulum



Superposition and Small Oscillations

Superposition of Two Harmonics

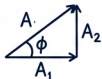
$$y_1(x, t) = A_1 \sin(kx - \omega t)$$

$$y_2(x, t) = A_2 \sin(kx - \omega t + \theta)$$

$$y_{\text{res}} = y_1 + y_2$$

Using phasor addition, $\Rightarrow A = \sqrt{A_1^2 + A_2^2 + 2 A_1 A_2 \cos \theta}$

Similarly the phase angle $\Rightarrow \tan \phi = \frac{A_2 \sin \theta}{A_1 + A_2 \cos \theta}$



Small Oscillations

Taylor expansion for a potential energy $U(x)$ around a stable equilibrium point x_0 :

$$U(x) \approx U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2} U''(x_0)(x - x_0)^2 + \dots$$

Since x_0 is an equilibrium, $U'(x_0) = 0$

For small displacements, force $\Rightarrow F \approx -U''(x_0)(x - x_0)$

This is an effective spring constant $k_{\text{eff}} = U''(x_0)$

$$\omega = \sqrt{\frac{U''(x_0)}{m}}$$

On general equation of motion for a small oscillation:

$$m\ddot{x} + U''(x_0)x = 0$$